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Unit 5 : Session 8 : SLO 2 | ARM CPSR, Modes and Software Interrupts

I. Fill in the Blanks

1. MRS → MRS Rd, CPSR moves the status register (including mode bits) into a general register.
2. Supervisor (SVC) → SWI triggers a software interrupt that enters Supervisor mode.
3. R8 → FIQ mode banks registers R8→R14.

II. True or False

1. TRUE → FIQ banks R8→R14 (7 registers) vs IRQ's R13→R14 (2), so FIQ handlers avoid stacking overhead.
2. FALSE → the CONTROL register belongs to ARMv7-M (Cortex-M); ARMv7-A switches modes via CPSR writes/exceptions.
3. TRUE → SWI is precisely the User → Supervisor transition mechanism (system calls).

III. Match the Following

1. SWI → 3. Software interrupt.
2. CPSR → 2. Holds processor status.
3. FIQ → 1. Fast interrupt mode.

IV. Numerical Answers

1. 5 → CPSR mode bits M[4:0] determine the processor mode.
2. 7 → FIQ banks R8_fiq through R14_fiq.
3. 2 → IRQ banks only R13_irq and R14_irq.